Curriculum Vitae/Resume

Donghoon Ahn
donghoon@berkeley.edu — Personal Page — LinkedIn — Lab page — GitHub

Research Interests

Deep Generative Models, **Diffusion Models**, **Image/Video/3D/4D Generation**, **Personalization**
Focused on integrating fundamental principles of the world into generative models as inductive biases to enhance image, video, and 3D model generation, with the aim of better understanding the real world.

EDUCATION

University of California, Berkeley 2025 — present
Ph.D. in Electrical Engineering and Computer Science
Advisor: Prof. Alexei A. Efros

Korea University 2019 — 2025
B.S. in Computer Science and Engineering GPA: 4.37/4.5 (overall) **4.46/4.5 (major)**
Two-year break for military service (Fall 2020 — Spring 2022)

WORK EXPERIENCES

Research Intern, Korea Advanced Institute of Science and Technology (KAIST) 2024 — 2025
Advised by Prof. Seungryong Kim
Currently researching diffusion models for video generation/personalization and analyzing and developing diffusion guidance methods.

Research Intern, Korea University 2022 — 2024
Advised by Prof. Seungryong Kim
Gained experience conducting independent research as an undergraduate intern over two years. Published papers at NeurIPS 2023 and ECCV 2024. Developed skills in defining research topics, designing experiments, coding, writing papers, and collaborating with peers.

Air Force Information System Management Unit 2020 — 2022
Refactored and developed several intranet systems as a team using the Spring framework during military service.

PUBLICATIONS

TAG: Tangential Amplifying Guidance for Hallucination-Resistant Diffusion Sampling
Hyunmin Cho*, **Donghoon Ahn***, Susung Hong*, Jee Eun Kim, Seungryong Kim, Kyong Hwan Jin
*International Conference on Machine Learning **(ICML)** 2026*

A Noise is Worth Diffusion Guidance
Donghoon Ahn*, Jiwon Kang*, Jaewon Min, Sanghyeon Lee, Wooseok Jang, Minjae Kim, Hyoungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha, Kyong Hwan Jin, Seungryong Kim
*The Fourteenth International Conference on Learning Representations **(ICLR)** 2026*

Video Camera Trajectory Editing with Generative Rendering from Estimated Geometry
Junyoung Seo*, Jisang Han*, Jaewoo Jung*, Siyoon Jin, JoungBin Lee, Takuya Narihira, Kazumi Fukuda, Takashi Shibuya, **Donghoon Ahn**, Shoukang Hu, Seungryong Kim, Yuki Mitsufuji
*Annual AAAI Conference on Artificial Intelligence **(AAAI)** 2026*

Where and How to Perturb: On the Design of Perturbation Guidance in Diffusion and Flow Models
Donghoon Ahn*, **Jiwon Kang***, Sanghyun Lee, Minjae Kim, Jaewon Min, Wooseok Jang, Saungwu Lee, Sayak Paul, Susung Hong, Seungryong Kim
*Advances in Neural Information Processing Systems **(NeurIPS)** 2025*

Identity-preserving Distillation Sampling by Fixed-Point Iterator
SeonHwa Kim, Jiwon Kim, Soobin Park, **Donghoon Ahn**, Jiwon Kang, Seungryong Kim, Kyong Hwan Jin, Eunju Cha
*Conference on Computer Vision & Pattern Recognition **(CVPR)** 2025*

Self-Rectifying Diffusion Sampling with Perturbed-Attention Guidance

Donghoon Ahn*, Hyoungwon Cho*, Jaewon Min, Wooseok Jang, Jungwoo Kim, SeonHwa Kim, Hyun Hee Park, Kyong Hwan Jin and Seungryong Kim
*European Conference on Computer Vision **(ECCV)** 2024*

Debiasing Scores and Prompts of 2D Diffusion for View-Consistent Text-to-3D Generation
Susung Hong*, **Donghoon Ahn*** (equal contribution), and Seungryong Kim
*Advances in Neural Information Processing Systems (**NeurIPS**) 2023*

PREPRINTS

Correction Training for Diffusion Models
Donghoon Ahn, Runqian Wang, Alexei A. Efros, Taesung Park, Yash Belhe
Arxiv preprint

Geometry-Aware Score Distillation via 3D Consistent Noising and Gradient Consistency Modeling
Min-Seop Kwak, **Donghoon Ahn**, Ines Hyeonsu Kim, Jin-Hwa Kim and Seungryong Kim
Arxiv preprint

SCHOLARSHIP AND AWARDS

Best Award, Independent Research Competition, Korea University — Spring 2024
1st place for research on"Self-Rectifying Diffusion Sampling with Perturbed-Attention Guidance"

Veritas Program Scholarship, Korea University — Spring 2024

Sung Ryun Scholarship Foundation Scholarship *(Full Tuition)* — Spring 2023, Spring 2024

Chang Gang Foundation Scholarship *(Full Tuition)* — Fall 2023

Semester High Honors, Korea University — Spring 2019, Spring 2020, Fall 2022, Spring 2023, Fall 2023

Dean's List, Korea University — Fall 2023
(GPA: 4.5/4.5)

Korea University Alumni Association Scholarship *(Full Tuition)* — Spring 2022

PROJECTS

Improving Diffusion-Based Image Restoration Models with Guidance — 2023 — 2024
*Collaborative research with **Samsung Electronics** to enhance camera technologies*
Conducted research to improve the performance of diffusion model-based inverse problem solvers by applying Self-Attention Guidance (SAG). Resolved challenges in applying SAG, including artifact issues in unguided regions, by introducing innovative solutions such as refined mask edges and an inverted mask. Explored various guidance methods.

Improving Disentanglement of Diffusion Models via Gaussian Mixture Attention Guidance — 2023
Course Project: 'Introduction to Computer Vision and Its Application'
Led a team to address object mixing issues in Stable Diffusion 1.x by following the latest research and defining key problems. Proposed a novel solution using Gaussian Mixture Models (GMM) for unsupervised learning applied to attention maps. Provided tutorials on diffusion models and contributed to coding and report writing.

SERVICE

Open Source Contribution, huggingface/diffusers — 2024
Collaborated with HuggingFace engineers to integrate Perturbed-Attention Guidance (PAG) into diffusers, significantly enhancing image/video generation quality.

AIKU: Deep Learning Society, *Korea University* — 2022 — 2023
Led a study group on generative models, focusing on diffusion models. Gave an introduction to image generation models and text-to-3D tasks.

SKILLS

- **Programming:** Python, PyTorch, C++, JAVA, Kotlin, Scala, SQL
- **Soft Skills:** Experienced in team collaboration and leadership through military service and research projects. Skilled in sharing and developing ideas with colleagues, and capable of leading research projects through task delegation and

role distribution.